Ramadan Mohamed Hassan

AI Engineer | Machine Learning | Computer Vision | LLMs | Full-Stack AI Applications

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PROFESSIONAL SUMMARY

Artificial Intelligence and Computer Science student focused on applied machine learning, computer vision, medical AI, smart agriculture, robotics, and full-stack AI applications. Huawei-certified AI trainee and Springer-published researcher with award-winning projects across healthcare AI, AgriTech, optimization algorithms, and data-driven software systems.

EDUCATION

New Mansoura University | B.Sc. Computer Science - Artificial Intelligence Program | Sep 2022 - Jun 2026

EXPERIENCE & TRAINING

ReNile Company - Artificial Intelligence Engineer Intern | *Mar 2026 - May 2026*

- Worked on deployment-oriented ML/DL workflows including data preprocessing, model optimization, and applied AI implementation.
- Collaborated with engineering/product stakeholders to translate requirements into practical AI-powered system components.

NVIDIA - AI & Prompt Engineering Trainee | *Aug 2025 - Oct 2025*

- Completed training in generative AI, LLM applications, prompt engineering, and AI-assisted workflow design.
- Strengthened Python foundations for data processing, prototyping, and end-to-end AI application development.

Huawei ICT Academy - HCIP AI Professional Intern | *Jul 2025 - Aug 2025*

- Completed professional-level AI training covering advanced machine learning concepts, AI architectures, and implementation workflows.

NTI / Huawei Egyptian Talent Academy - Artificial Intelligence Trainee | *Jul 2025 - Aug 2025*

- Completed an 80-hour AI training program with a 97% score, covering ML algorithms, data analysis, and model evaluation.

Huawei ICT Academy - HCIA AI Intern | *Jun 2025 - Jul 2025*

- Built foundations in industry AI concepts, data preparation, algorithm selection, and applied machine learning.

RESEARCH & PUBLICATIONS

- **Springer Conference Proceedings - ISBCOM 2025:** "Hybrid Random Walk-Dijkstra Approach for Efficient Coverage Path Planning in Robotics." Designed a robotic path-planning approach combining Random Walk, Dijkstra's algorithm, and computer vision for obstacle-aware navigation. Publication

SELECTED TECHNICAL PROJECTS

- **AgriNova - AI, Robotics, IoT & Blockchain Smart Agriculture System:** Flagship smart agriculture platform combining AI crop intelligence, VGG16 plant disease detection (96.9% accuracy), IoT-style monitoring, UAV/drone workflows, robotics, blockchain-inspired traceability, and interactive analytics dashboards. 1st Place CSE Projects Exhibition 2026 and 2nd Place MansTech Hackathon with 30,000 EGP prize. GitHub
- **Brain Tumor MRI AI System:** CNN-based medical imaging pipeline for automated brain tumor classification from MRI scans using TensorFlow/Keras, preprocessing, model training, and evaluation workflows. GitHub
- **AI MedScan Pro:** Streamlit medical AI app for image analysis, retinal analysis, and disease-risk prediction using Scikit-learn, OpenCV/Pillow, and interactive reporting. GitHub
- **Gastrointestinal Tract Anomaly Detection:** Applied Faster R-CNN Inception V2 on the KVASIR endoscopy dataset for anomaly detection, reporting 89.2% average precision and 0.28s inference time. GitHub
- **Molecular Property Prediction with GNN:** Graph Neural Network project for molecular representation learning and property prediction from graph-structured chemical data. GitHub
- **AI Nutritional Recommendation System:** TabNet-based recommendation system analyzing 26 infant nutrition/macronutrient features and reporting 99% accuracy for personalized diet suggestions. GitHub
- **Diabetes Prediction ML Pipeline:** End-to-end medical ML pipeline with EDA, preprocessing, outlier handling, feature engineering, model comparison, and XGBoost/LightGBM evaluation, reporting 98% accuracy. GitHub
- **Breast Cancer Prediction System:** Supervised ML classification workflow with preprocessing, model comparison, class balancing, and interactive visualization, reporting 92% accuracy for early-risk analysis. GitHub
- **Hybrid Optimization Algorithm:** Award-winning optimization project combining metaheuristic search strategies and benchmarking on standard optimization functions. GitHub
- **FinLit Egypt Financial Literacy Platform:** Full-stack financial literacy platform with educational content, calculators, user flows, saved calculations, and Express/JavaScript backend. GitHub
- **Exoplanet Detection ML System:** Scientific ML pipeline for detecting planetary transits in noisy light-curve data with model comparison and reproducible evaluation. GitHub
- **NLP Text Classification with BERT:** Transformer-based text classification project using BERT-style NLP workflows for sentiment/text classification, reporting 94% accuracy on benchmark datasets. GitHub

AWARDS & CERTIFICATIONS

- 2nd Place - MansTech Hackathon (Apr 2026): Won 30,000 EGP for an AI, robotics, and IoT agricultural solution endorsed by ITIDA.
- 1st Place - Computer Science and Engineering Projects Exhibition 2026: Awarded for AgriNova smart agriculture project.
- 1st Place - Annual University Projects Exhibition 2025: Recognized for an innovative optimization algorithm project among 45 projects.
- Springer Publication Certificate (2025): Published in Springer conference proceedings for robotic path planning research.
- ICPC & ECPC Qualification (2023): Ranked in the top 15% of teams according to CV evidence.
- NVIDIA: Building LLM Applications with Prompt Engineering; AI for All: From Basics to GenAI Practice.
- Huawei ICT Academy: HCIA AI, HCIP AI; NTI/Huawei Egyptian Talent Academy AI Training - 97% score.

TECHNICAL SKILLS

Programming: Python, JavaScript, TypeScript, C, C++, C#, SQL, HTML5, CSS3
AI/ML: Machine Learning, Deep Learning, Predictive Modeling, Feature Engineering, Model Evaluation, Transfer Learning
Frameworks: TensorFlow, Keras, PyTorch, Scikit-learn, XGBoost, LightGBM, Pandas, NumPy, Matplotlib, Seaborn, Plotly
Computer Vision & NLP: OpenCV, CNNs, Faster R-CNN, VGG16, BERT, Transformers, LLMs, Generative AI, Prompt Engineering
Web & Backend: React, Next.js, Node.js, Express.js, REST APIs, Streamlit, Responsive Design, Recharts
Data / Cloud / Tools: PostgreSQL, MongoDB, ETL, Git, GitHub, AWS, GCP, Vercel, Hugging Face, LangChain, RAG, FAISS, Pinecone
Specialized Areas: Graph Neural Networks, Optimization Algorithms, Medical Image Analysis, Robotics, IoT, UAV workflows, Blockchain concepts

COURSES & LANGUAGES

Courses: Computer Vision Specialization - Coursera; NLP with Deep Learning - Coursera; Full Stack Web Development - Creativo; Introduction to Data Science - Udemy; Introduction to AI - Udemy.
Languages: Arabic - Native; English - Professional Working Proficiency.